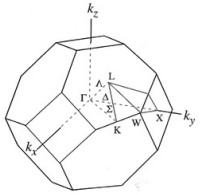
Carrier Density

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Carrier Density

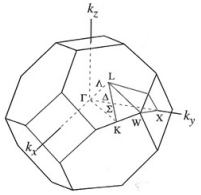
- Pauli exclusion principle
  - Only one electron can occupy a given quantum state
  - Carrier density = number of state x probability of occupation
- Only a fraction of states are occupied
- How many states are occupied up to E?

Or equivalently…

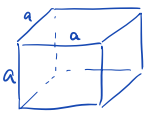
- How many states per unit energy? (DOS)

Density of state (DOS). $\Rightarrow D(E)$ $D(E)\cdot dE$: number of state in $E \sim E+dE$

number of states per unit energy.



$$V(x,y,z) = \begin{cases} 0, & 0<x<a \\ & 0<y<a \\ & 0<z<a \\ \infty, & \text{other} \end{cases}$$



$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right)+V\cdot\psi = E\cdot\psi$$

$0<x<a$, $0<y<a$, $0<z<a$: $-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}\right) = E\cdot\psi(x,y,z)$

other : $\psi(x,y,z)=0$

$\psi(x,y,z) = \psi(x)\cdot\psi(y)\cdot\psi(z)$

$$-\frac{\hbar^2}{2m}\left(\psi_x''\psi_y\psi_z+\psi_y''\psi_x\psi_z+\psi_z''\psi_x\psi_y\right) = E\,\psi_x\psi_y\psi_z$$

$$-\frac{\hbar^2}{2m}\left(\frac{\psi_x''}{\psi_x}+\frac{\psi_y''}{\psi_y}+\frac{\psi_z''}{\psi_z}\right) = E$$

E_x E_y E_z Constant

$$-\frac{\hbar^2}{2m}\cdot\frac{\psi_x''}{\psi_x} = E_x$$

$$E_x = \frac{\hbar^2 k_x^2}{2m} \qquad k_x = \frac{n_x\pi}{a}$$

$$E = \frac{\hbar^2}{2m}\left[k_x^2+k_y^2+k_z^2\right] = \frac{\hbar^2}{2m}\left(\frac{\pi}{a}\right)^2\left[n_x^2+n_y^2+n_z^2\right]$$

$$(E, E+dE) \longrightarrow \frac{1}{8}\cdot 4\pi k^2\,dk$$

$$D(E)\cdot dE = \frac{\frac{1}{8}\cdot 4\pi k^2\,dk}{\left(\frac{\pi}{a}\right)^3}$$

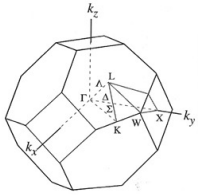
$$D(E) = \frac{\frac{1}{8}\cdot 4\pi k^2}{\left(\frac{\pi}{a}\right)^3}\cdot\frac{dk}{dE}$$

reduced DOS : $g(E) = \frac{D(E)}{a^3} = \frac{\frac{1}{8}\cdot 4\pi\cdot k^2}{\pi^3}\cdot\frac{dk}{dE}$

$E = \frac{\hbar^2 k^2}{2m}$ $\Rightarrow$ $g(E) = \frac{1}{4\pi^2}\cdot\left(\frac{\sqrt{2m}}{\hbar}\right)^3\cdot\sqrt{E} \propto \sqrt{E}$

Density of State

- Mathematical Derivation
- Consider a free electron confined to a 3D infinite potential well
  - $V(x, y, z) = 0$ for 0<x<a, 0<y<a, 0<z<a
  - $V(x, y, z) = \infty$ elsewhere
- What is relationship between energy and wavenumber k?
- What wavenumber k looks like?



Density of State

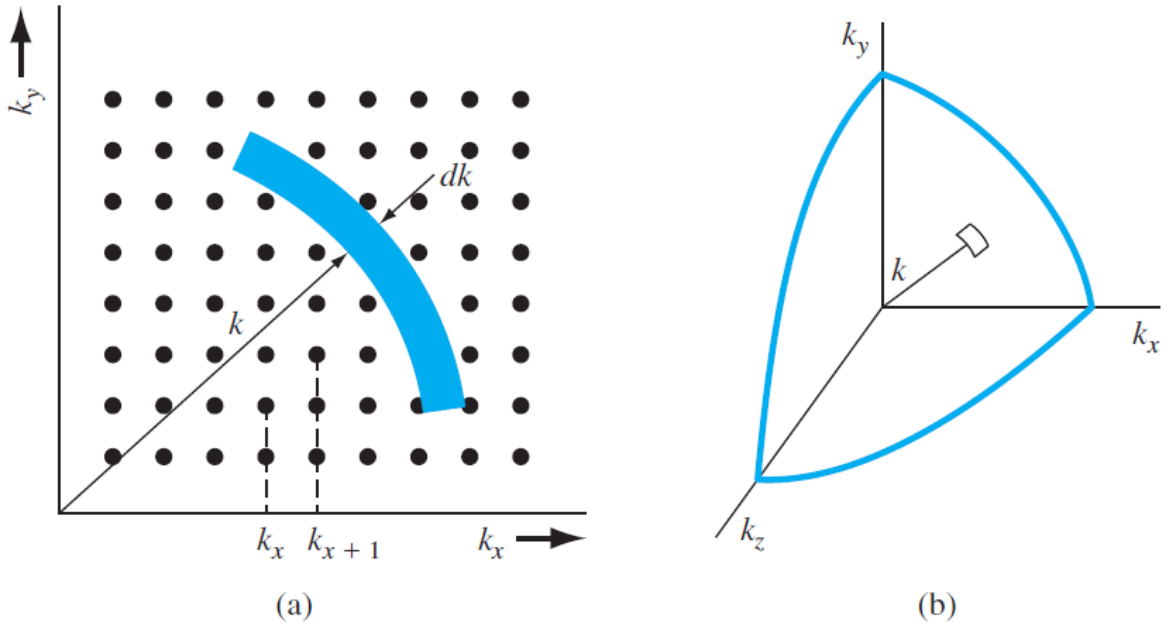
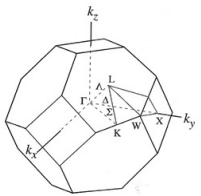
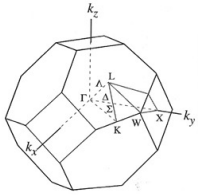


Figure 3.26 | (a) A two-dimensional array of allowed quantum states in k space. (b) The positive one-eighth of the spherical k space.



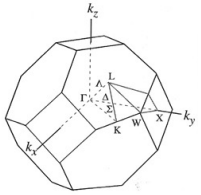
Density of State in k Space

- This distance between two quantum states in the k_x direction is given by $k_{x+1} - k_x = (n_x + 1)\left(\frac{\pi}{a}\right) - (n_x)\left(\frac{\pi}{a}\right) = \left(\frac{\pi}{a}\right)$
- The volume V_k of a single quantum state is $V_k = \frac{1}{2} \times \left(\frac{\pi}{a}\right)^3$
  - Note: $\frac{1}{2}$ is for spin.
- The differential volume in k space is given by $\frac{1}{8} \times 4\pi k^2 dk$
- The differential density of quantum states in k space can be written as
  - $g_T(k)dk = \frac{\frac{1}{8} \times 4\pi k^2 dk}{\frac{1}{2} \times \left(\frac{\pi}{a}\right)^3} = g_T(E)dE$



Density of State as a Function of Energy

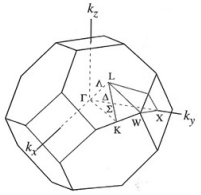
- $g_T(k)dk = \dfrac{\frac{1}{8}\times 4\pi k^2 dk}{\frac{1}{2}\times\left(\frac{\pi}{a}\right)^3} = g_T(E)dE$
- Using the relationship: $E = \dfrac{\hbar^2 k^2}{2m}$
  - $g_T(E)dE = \dfrac{4\pi a^3}{h^3}\cdot(2m)^{3/2}\cdot\sqrt{E}dE$
    - ✓ Total number of quantum states between the energy E and E+dE in the crystal space volume of a^3
  - $g(E) = \dfrac{4\pi}{h^3}\cdot(2m)^{3/2}\cdot\sqrt{E}$
    - ✓ The density of states per unit volume of crystal



Extension to Semiconductors

- The E versus k near k=0 at the bottom of the conduction band:
  - $E = E_c + \frac{\hbar^2 k^2}{2m_n^*}$
  - $E - E_c = \frac{\hbar^2 k^2}{2m_n^*}$
  - $g_c(E) = \frac{4\pi}{h^3} \cdot (2m_n^*)^{3/2} \cdot \sqrt{E - E_c}$ valid for E>E_c
- The E versus k near k=0 at the top of the valence band:
  - $E = E_v - \frac{\hbar^2 k^2}{2m_p^*}$
  - $E_v - E = \frac{\hbar^2 k^2}{2m_p^*}$
  - $g_v(E) = \frac{4\pi}{h^3} \cdot \left(2m_p^*\right)^{3/2} \cdot \sqrt{E_v - E}$ valid for E<E_v

Questions: what are the values of m_n^* and m_p^* ? Are they just effective mass for all semiconductors?



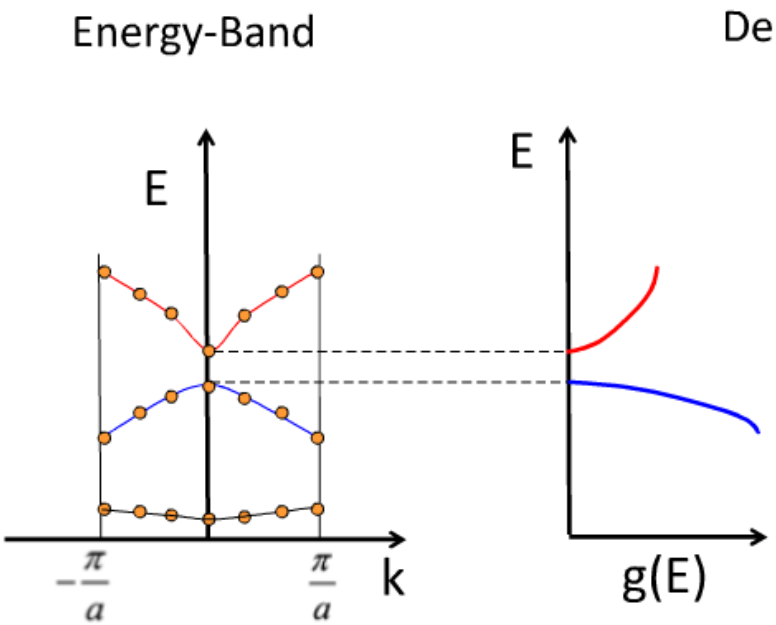
heavy hole: $g_{hh}(E) = \frac{4\pi}{h^3}(2m_{hh}^*)^{\frac{3}{2}}\cdot\sqrt{E_v - E}$

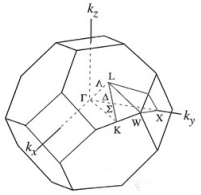
light hole: $g_{lh}(E) = \frac{4\pi}{h^3}(2m_{lh}^*)^{\frac{3}{2}}\cdot\sqrt{E_v - E}$

$$g_v(E) = g_{hh}(E) + g_{lh}(E) = \frac{4\pi}{h^3}(2m_v^*)^{\frac{3}{2}}\cdot\sqrt{E_v - E}$$

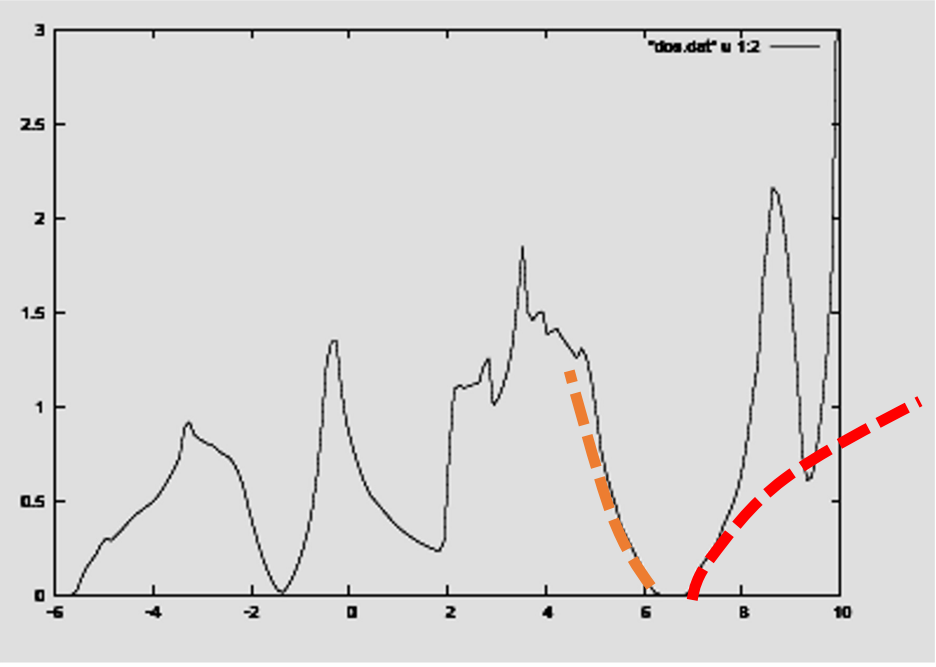
$$\Rightarrow m_{hh}^{*\frac{3}{2}} + m_{lh}^{*\frac{3}{2}} = m_v^{*\frac{3}{2}}$$

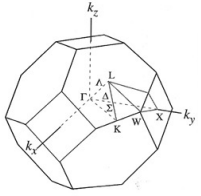
E-k diagram and Density of State



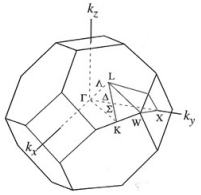


Real DOS needs computation



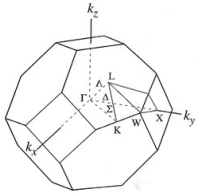


But How Do We Fill These States?



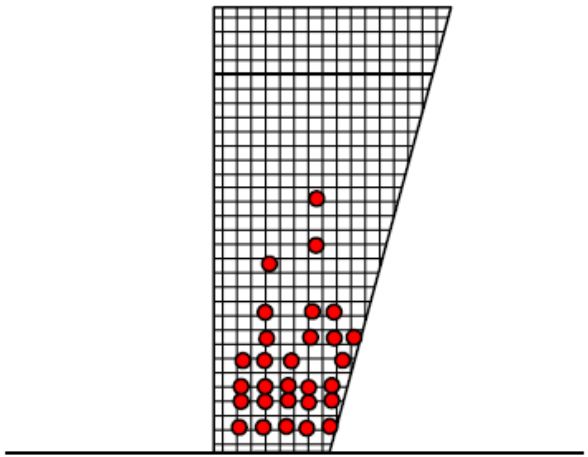
Fermi-Dirac Statistics

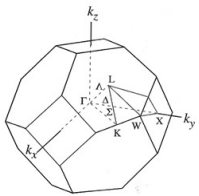
电子 ⟶ 费米子：一个态上只有一个电子



Carrier Density

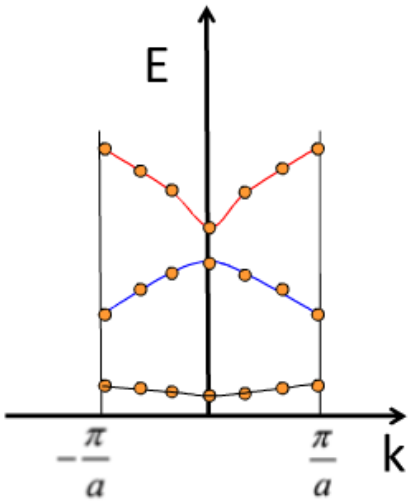
- Carrier number = Number of States x filling factor



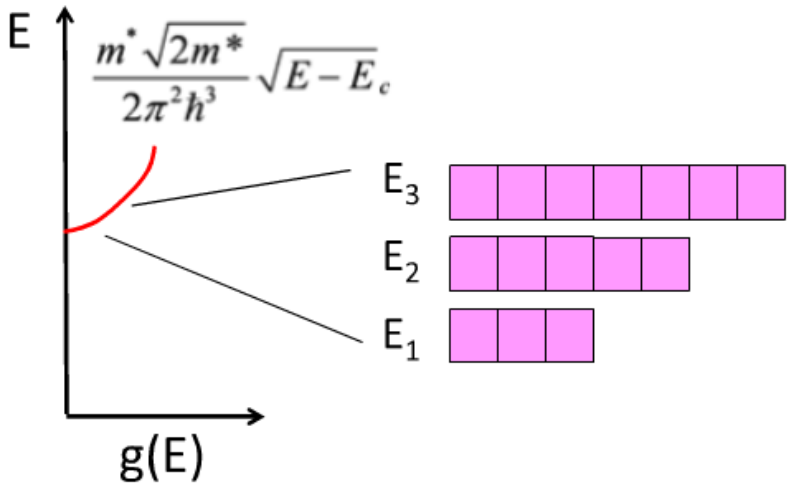


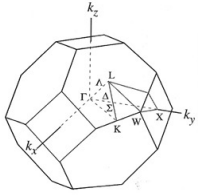
E-k diagram and Electronic States

Energy-Band

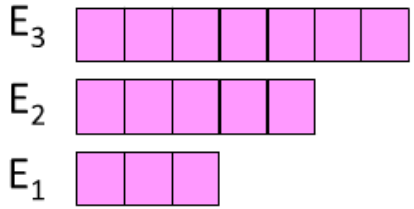


Density of States



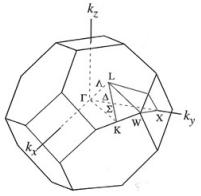


Rules for Filling Up the States



Consider a system with fixed number of electrons and total fixed energy

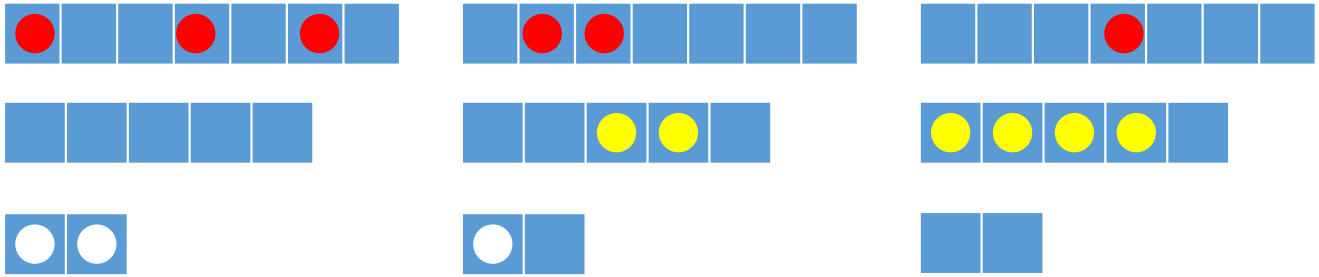
- Pauli principle: Only one electron per state(ignore spin at this time)
- Total number of electron is conserved: $N_t = \sum N_i$
- Total energy of the system is conserved: $E_t = \sum E_i N_i$



Illustrative Example: 3 Energy Levels

- $N_t = 5$ and $E_t = 12$, $N_t = \sum N_i \;\; E_t = \sum E_i N_i$

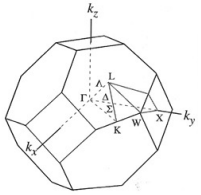
- E=4
  E=2
  E=0



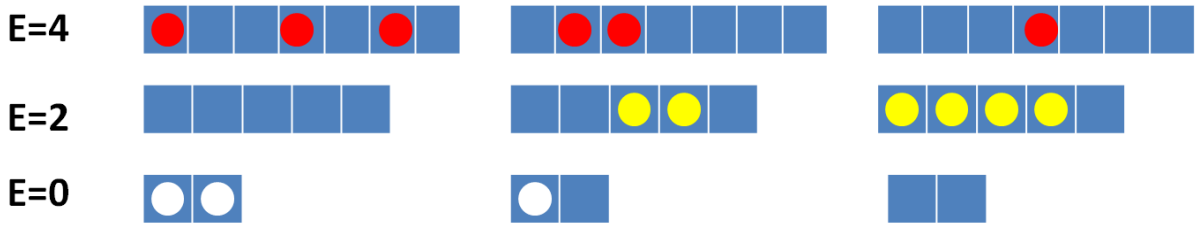
$$W_{203} = \frac{2!}{1!2!} \bullet \frac{5!}{0!5!} \bullet \frac{7!}{3!4!} = 35$$

$$W_{122} = \frac{2!}{1!1!} \bullet \frac{5!}{2!3!} \bullet \frac{7!}{5!2!} = 420$$

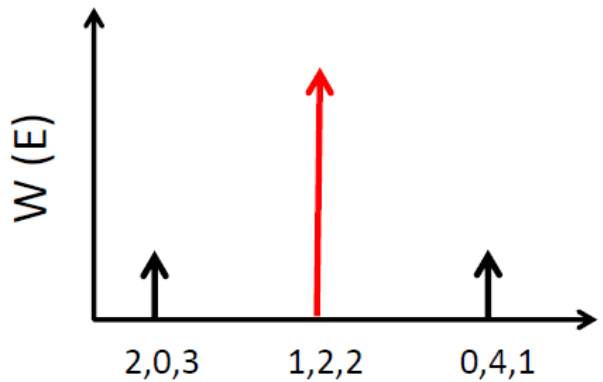
$$W_{041} = \frac{2!}{0!2!} \bullet \frac{5!}{4!1!} \bullet \frac{7!}{6!1!} = 35$$

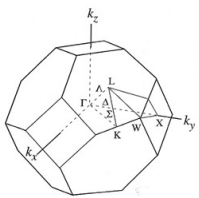


Occupation Statistics



Choose the most probable configuration



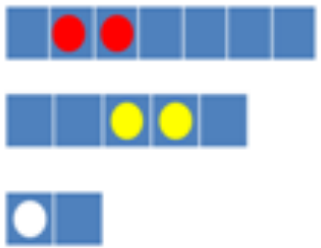


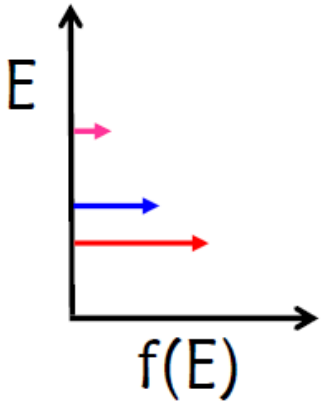
Occupation Statistics

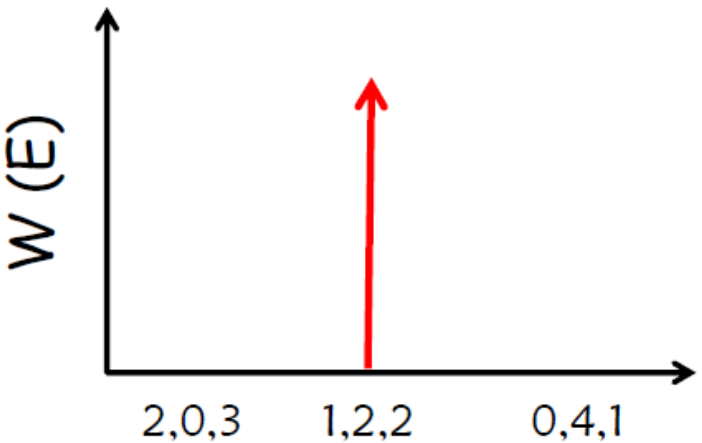
E=4, f_3=2/7

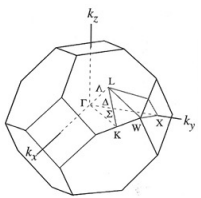
E=2, f_2= 2/5

E=0, f_1 =1/2

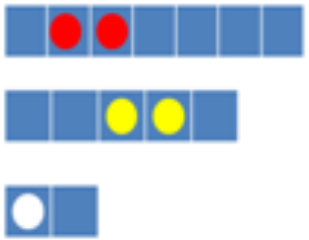


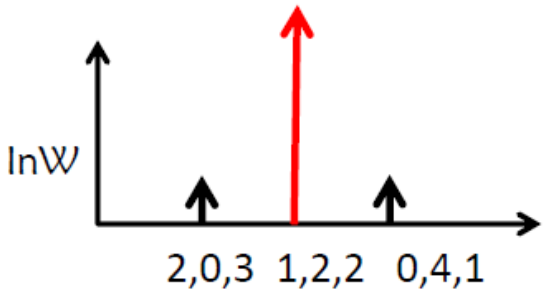






For N-states





$$W_c = \prod_i \frac{S_i!}{(S_i - N_i)!N_i!}$$

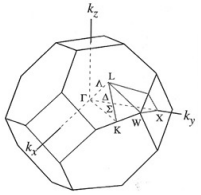
Recall. $W_{203} = \frac{2!}{1!2!} \bullet \frac{5!}{0!5!} \bullet \frac{7!}{3!4!}$

$$ln(W_c) = \sum_i [lnS_i! - \ln(S_i - N_i)! - lnN_i!]$$

Stirling's Approximation

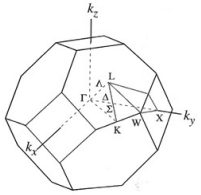
$$\cong \sum_i [S_i lnS_i - S_i - (S_i - N_i)\ln(S_i - N_i) + (S_i - N_i) - N_i lnN_i + N_i]$$

$$= \sum_i [S_i lnS_i - (S_i - N_i)\ln(S_i - N_i) - N_i lnN_i]$$



Optimization with Lagrange-Multiplier

- Consider the optimization problem:
  - Maximize f(x, y), subject to g(x, y) = 0
- Define Lagrange function
  - $\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda \cdot g(x, y)$
- Solve:
  - $\nabla_{x,y,\lambda}\mathcal{L}(x, y, \lambda) = 0$

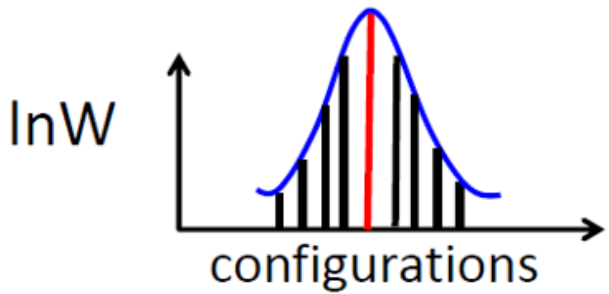


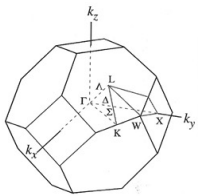
Optimization with Lagrange-Multiplier

- $\delta \ln(W_c) = \sum_i \frac{\partial ln W_c}{\partial N_i} dN_i$
- $\cong \sum_i \left[\ln\left(\frac{S_i}{N_i} - 1\right) dN_i\right] - \alpha \sum_i dN_i - \beta \sum_i E_i dN_i$
- $= \sum_i \left[\ln\left(\frac{S_i}{N_i} - 1\right) - \alpha - \beta E_i\right] dN_i$

$E_T = \sum_i E_i N_i$

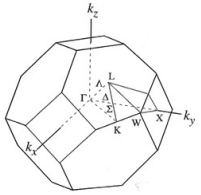
$N_T = \sum_i N_i$





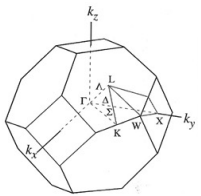
Final Step

- $\ln\left(\frac{S_i}{N_i}-1\right)-\alpha-\beta E_i=0$
- $f(E)\equiv\frac{N_i}{S_i}=\frac{1}{1+e^{\alpha+\beta E_i}}$
- $f_{max}(E)$=1
- What is α and β?



Final Step

- $\ln\left(\frac{S_i}{N_i}-1\right)-\alpha-\beta E_i=0$
- $f(E_i)\equiv\frac{N_i}{S_i}=\frac{1}{1+e^{\alpha+\beta E_i}}$ $f(E)=\frac{1}{1+e^{\alpha+\beta E}}$
- $f_{max}(E)$=1
- At $E = E_F, f(E_F)\equiv\frac{1}{2},$ $\alpha+\beta E_F=0$
- $f(E) = \frac{1}{1+e^{\beta(E-E_F)}}$
- At $E\to\infty$, it will converge to Boltzmann distribution
  - $f_{Boltzmann}(E)=Ae^{-E/k_BT}$
  - $\beta=\frac{1}{k_BT}$



Fermi-Dirac Probability Function

When $E-E_f>>kT$:

$$f_F(E) \approx e^{-(E-E_F)/k_BT}$$

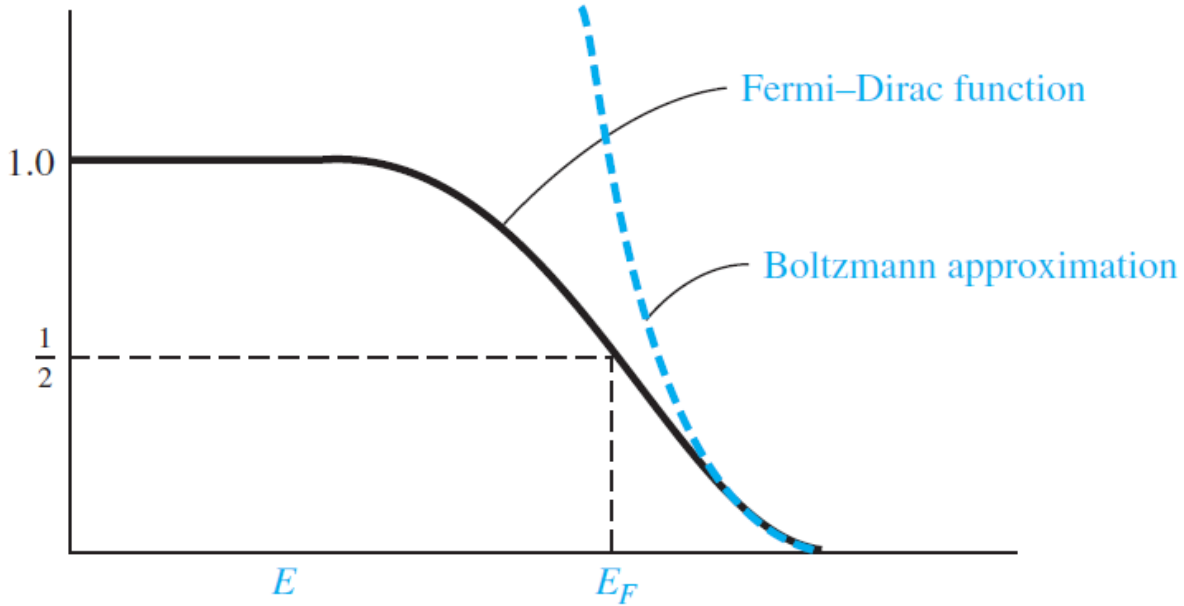
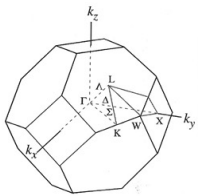
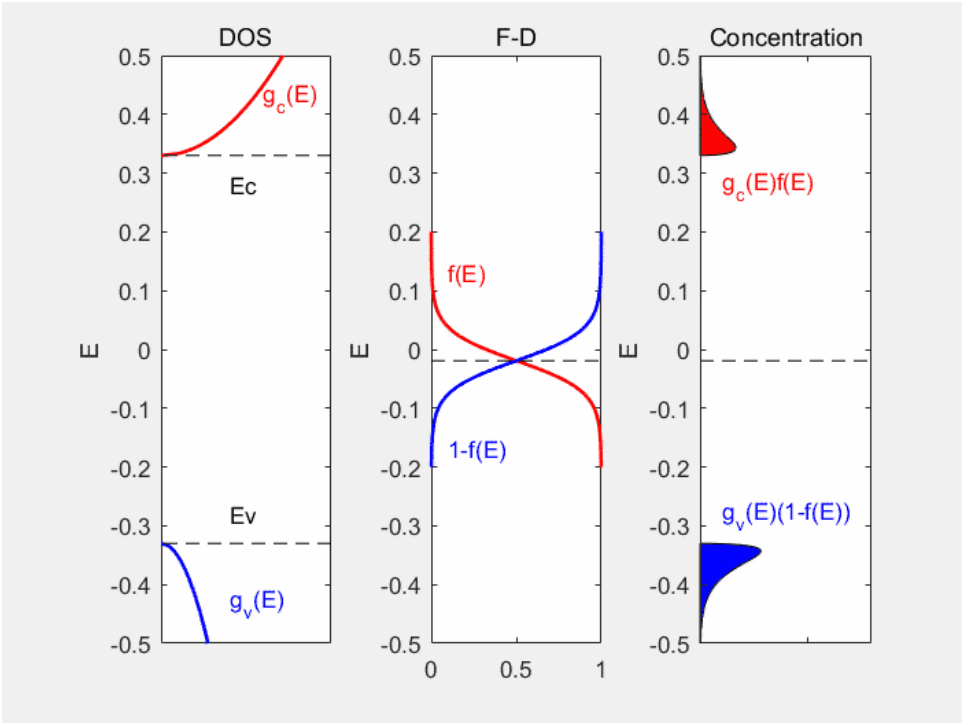


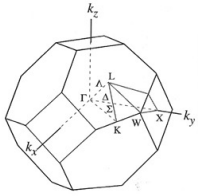
Figure 3.35 | The Fermi–Dirac probability function and the Maxwell–Boltzmann approximation.



Carrier Distribution

- $n(E) = g_c(E)f_F(E)$
- $p(E) = g_v(E)[1 - f_F(E)]$

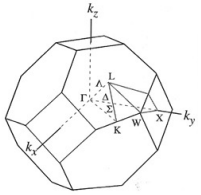




Electron Concentration

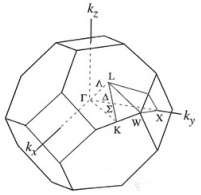
- $n_0 = \int n(E)dE = \int g_c(E)f_F(E)\,dE = \int_{E_c}^{\infty} g_c(E)f_F(E)\,dE$
- If Fermi level is within the forbidden-energy band gap,
  - $E_c - E_F \gg kT$
  - $f_F(E) = \frac{1}{1+\exp(E-E_F)/kT} \approx \exp[\frac{-(E-E_F)}{kT}]$
  - $n_0 = \int_{E_c}^{\infty} \frac{4\pi}{h^3} \cdot (2m_n^*)^{3/2} \cdot \sqrt{E-E_c}\exp[\frac{-(E-E_F)}{kT}]\,dE$
  - $n_0 = \frac{4\pi}{h^3} \cdot (2m_n^* kT)^{3/2}\exp[\frac{-(E_c-E_F)}{kT}] \int_0^{\infty} \sqrt{\eta}\exp(-\eta)d\eta$
  - $n_0 = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} \exp\left[\frac{-(E_c-E_F)}{kT}\right] = N_c \exp\left[\frac{-(E_c-E_F)}{kT}\right]$

$\frac{1}{2}\sqrt{\pi}$



Hole Concentration

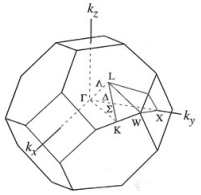
- $p_0 = \int p(E)dE = \int g_v(E)(1 - f_F(E))\, dE$
- If Fermi level is within the forbidden-energy band gap,
  - $E_F - Ev \gg kT$
  - $1 - f_F(E) = 1 - \frac{1}{1+\exp(E-E_F)/kT} \approx \exp[\frac{-(E_F-E)}{kT}]$
  - $p_0 = \int_{\infty}^{E_v} \frac{4\pi}{h^3} \cdot (2m_p^*)^{\frac{3}{2}} \cdot \sqrt{E_v - E} \exp[\frac{-(E_F-E)}{kT}]\, dE$
  - $p_0 = -\frac{4\pi}{h^3} \cdot (2m_p^* kT)^{3/2} \exp[\frac{-(E_F-E_v)}{kT}] \int_{\infty}^{0} \sqrt{\eta} \exp(-\eta) d\eta$
  - $p_0 = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \exp[\frac{-(E_F-E_v)}{kT}] = N_v \exp[\frac{-(E_F-E_v)}{kT}]$



Example

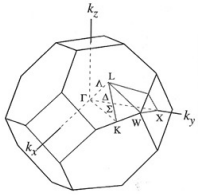
Table 4.1 | Effective density of states function and density of states effective mass values

	N_c (cm^{-3})	N_v (cm^{-3})	m_n^*/m_0	m_p^*/m_0
Silicon	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37



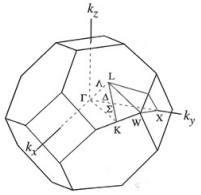
Intrinsic Carrier Concentration

- For an intrinsic semiconductor, the concentration of electrons in the conduction band is equal to the concentration of holes in the valence band
- $n_i = p_i$
- The Fermi energy level for intrinsic semiconductor is called the intrinsic Fermi energy: $E_F = E_{Fi}$
- $n_0 = n_i = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$, $p_0 = p_i = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right]$
- $n_i^2 = N_c N_v \exp\left[\frac{-(E_c - E_v)}{kT}\right] = N_c N_v \exp\left[\frac{-(E_g)}{kT}\right]$



A Few Comments

- It is a very strong function of temperature and material *bandgap*
- *Where is the intrinsic Fermi-level?*
  - ➢ *Almost pin to midgap energy* $\frac{1}{2}(E_c + Ev)$
  - ➢ *Why not exactly pin to midgap energy?*



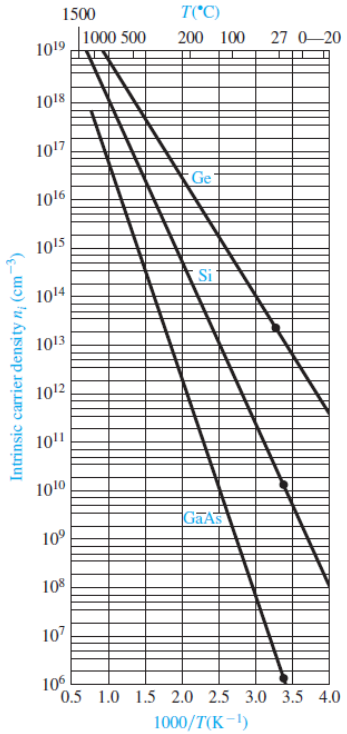


Figure 4.2 | The intrinsic carrier concentration of Ge, Si, and GaAs as a function of temperature. *(From Sze [14].)*

Dopant Atoms and Energy Levels

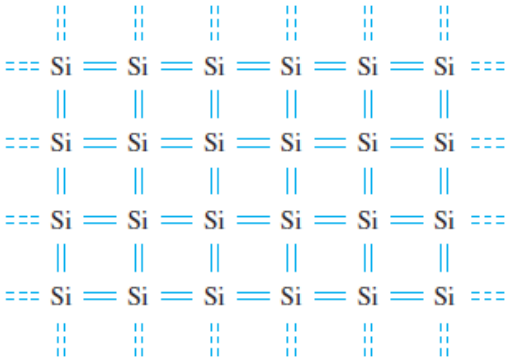


Figure 4.3 | Two-dimensional representation of the intrinsic silicon lattice.

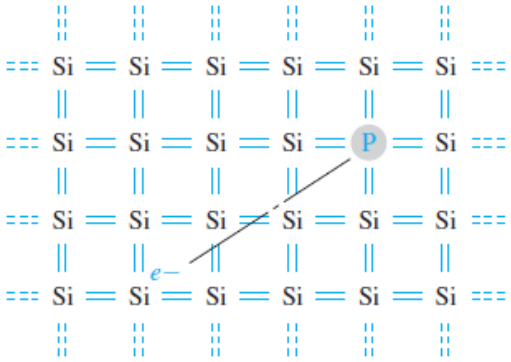
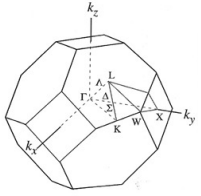


Figure 4.4 | Two-dimensional representation of the silicon lattice doped with a phosphorus atom.



Dopant Atoms and Energy Levels

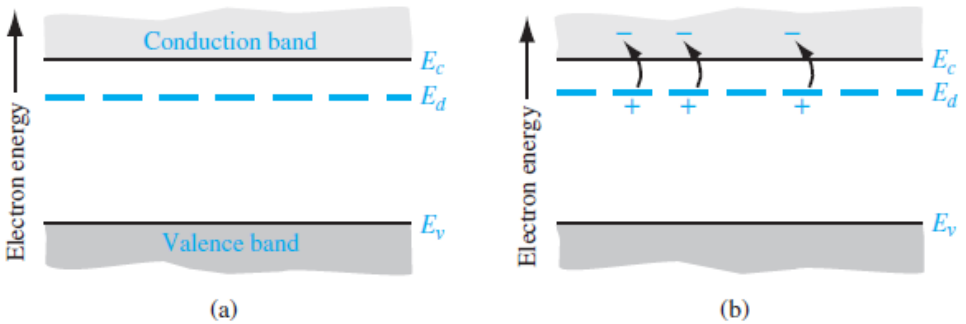
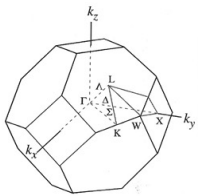


Figure 4.5 | The energy-band diagram showing (a) the discrete donor energy state and (b) the effect of a donor state being ionized.



Dopant Atoms and Energy Levels

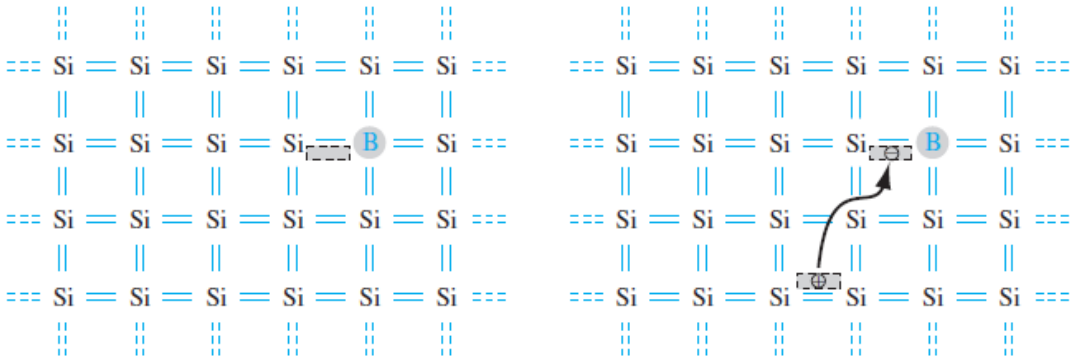
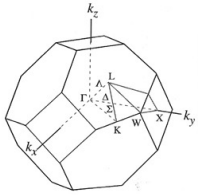


Figure 4.6 | Two-dimensional representation of a silicon lattice (a) doped with a boron atom and (b) showing the ionization of the boron atom resulting in a hole.



Dopant Atoms and Energy Levels

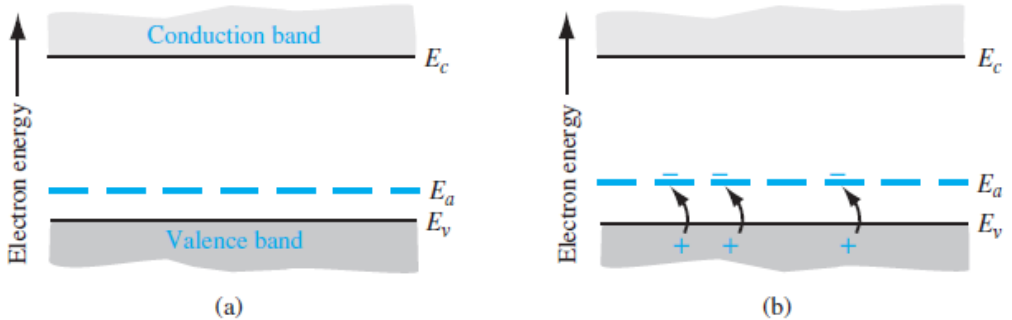
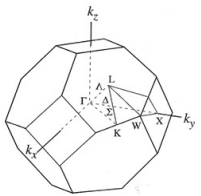


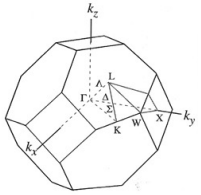
Figure 4.7 | The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.



Impurity Ionization Energies

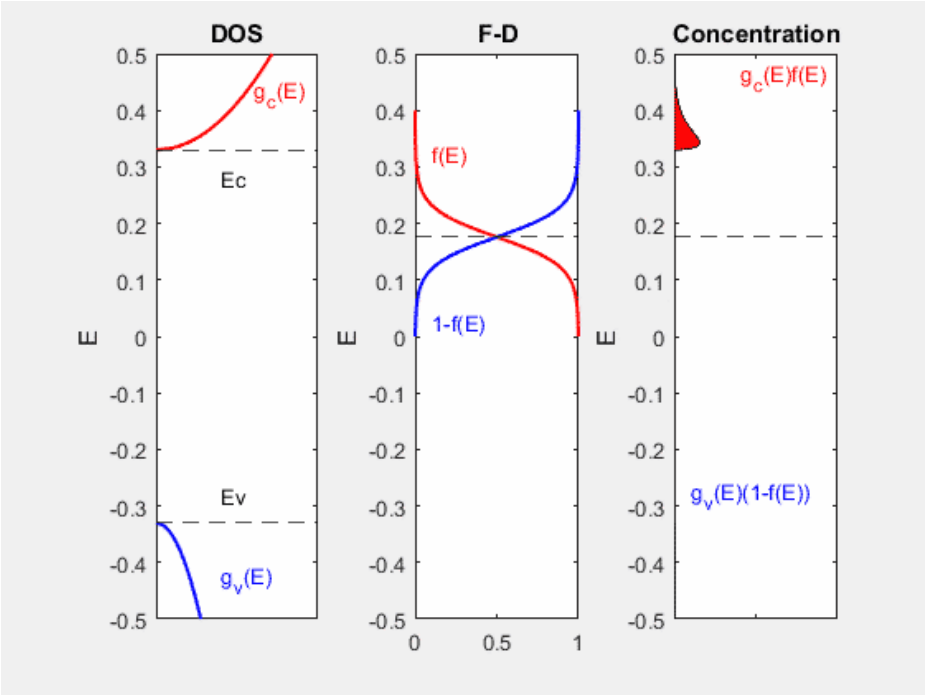
Table 4.3 | Impurity ionization energies in silicon and germanium

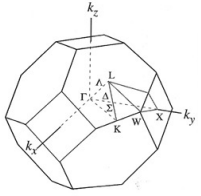
Impurity	Ionization energy (eV) Si	Ge
Donors		
Phosphorus	0.045	0.012
Arsenic	0.05	0.0127
Acceptors		
Boron	0.045	0.0104
Aluminum	0.06	0.0102



The Extrinsic Semiconductor

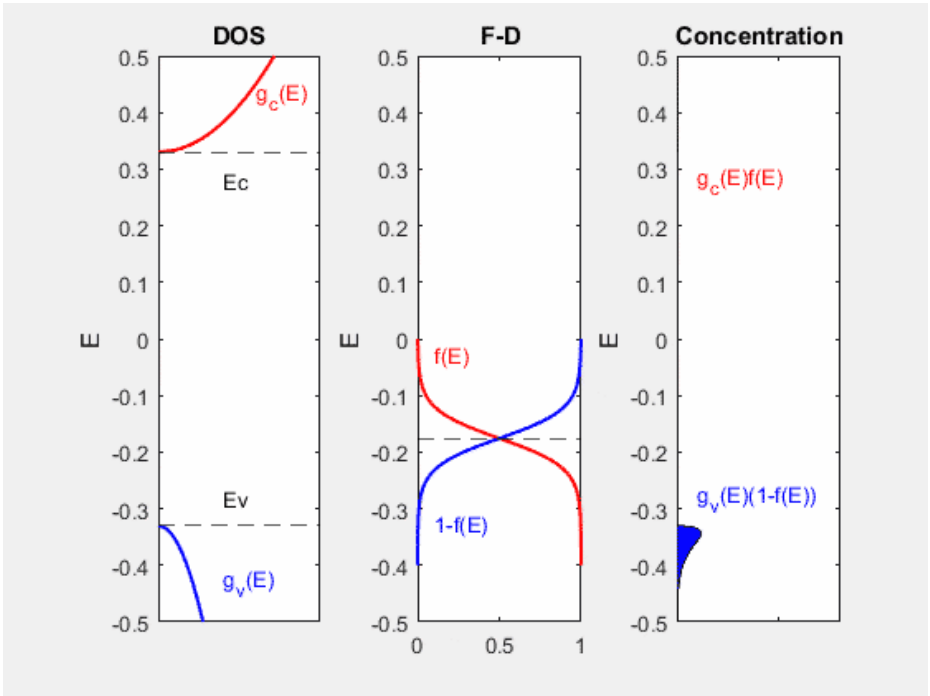
- N-type Semiconductor

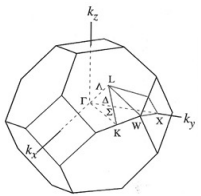




The Extrinsic Semiconductor

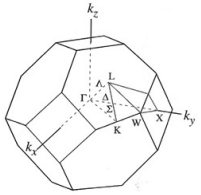
- P-type Semiconductor





Another Form of Electrons and Holes

- $n_0 = n_i \exp\left[\frac{(E_F - E_{Fi})}{kT}\right]$
- $p_0 = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]$
- $n_0 p_0 = n_i^2$



Degenerate Semiconductors

Boltzmann Approximation was not valid anymore.

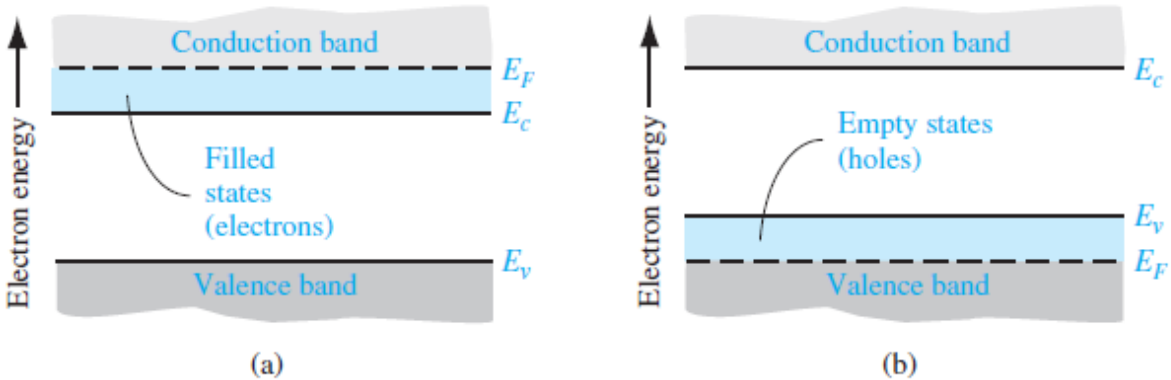
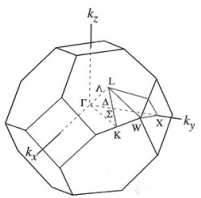
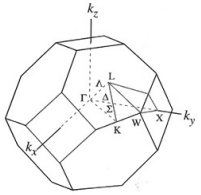


Figure 4.11 | Simplified energy-band diagrams for degenerately doped (a) n-type and (b) p-type semiconductors.



Degenerate Semiconductors

- **Boltzmann Approximation was not valid anymore.**
  - $f_F(E) = \frac{1}{1+\exp(E-E_F)/kT} \approx \exp[\frac{-(E-E_F)}{kT}]$
  - $n_0 = \int_{E_c}^{\infty} \frac{4\pi}{h^3} \cdot (2m_n^*)^{3/2} \cdot \sqrt{E-E_c} \frac{1}{1+\exp(E-E_F)/kT} dE$
  - Define: $\eta = \frac{(E-E_c)}{kT}$, $\eta_F = \frac{(E_F-E_c)}{kT}$
  - $n_0 = \frac{4\pi}{h^3} \cdot (2m_n^* kT)^{3/2} \int_0^{\infty} \frac{\eta^{1/2}}{1+\exp(\eta-\eta_F)} d\eta = \frac{4\pi}{h^3} \cdot (2m_n^* kT)^{\frac{3}{2}} F_{\frac{1}{2}}(\eta_F)$
- Similarly:
  - $\eta' = \frac{(E_v-E)}{kT}$, $\eta'_F = \frac{(E_v-E_F)}{kT}$
  - $p_0 = \frac{4\pi}{h^3} \cdot (2m_p^* kT)^{3/2} \int_0^{\infty} \frac{\eta'^{1/2}}{1+\exp(\eta'-\eta'_F)} d\eta = \frac{4\pi}{h^3} \cdot (2m_p^* kT)^{\frac{3}{2}} F_{\frac{1}{2}}(\eta'_F)$



The Fermi-Dirac Integral $F_{1/2}$

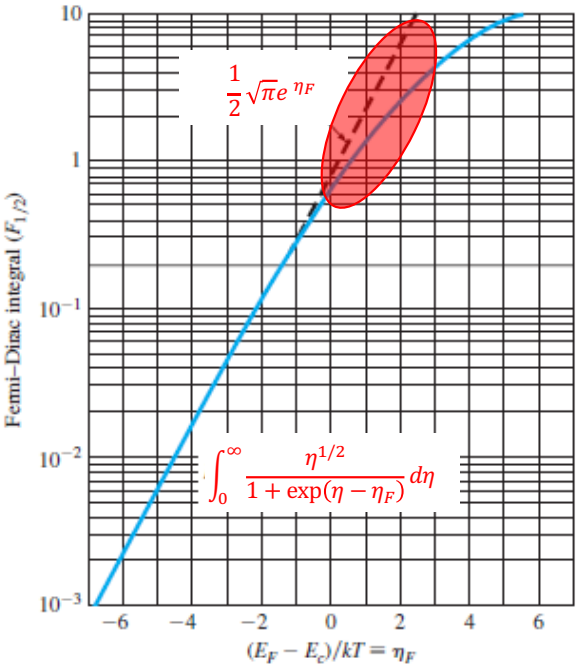
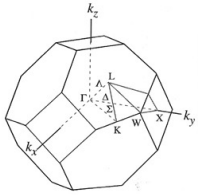
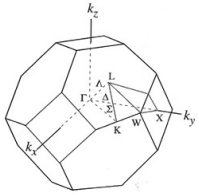


Figure 4.10 | The Fermi–Dirac integral $F_{1/2}$ as a function of the Fermi energy.

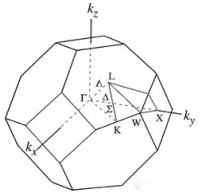


Statistics of Donors and Acceptors



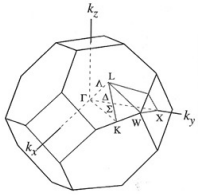
Probability Function

- The probability function of electrons occupying the donor state is
  - $\frac{n_d}{N_d} = \frac{1}{1+\frac{1}{g}\exp(\frac{E_d - E_F}{kT})}$
  - g is called degeneracy factor, is typically 2 for electron, due to the spin
  - n_d is the density of electrons occupying the donor level
  - $N_d^+ = N_d - n_d$: the concentration of ionized donors. We are more interested in this quantity.
- The probability function of hole occupying the acceptor state is
  - $\frac{p_a}{N_a} = \frac{1}{1+\frac{1}{g}\exp(\frac{E_F - E_a}{kT})}$
  - g is typically 4 for acceptor level in silicon and GaAs
  - p_a is the density of holes occupying the acceptor level
  - $N_a^- = N_a - p_a$: the concentration of ionized acceptors

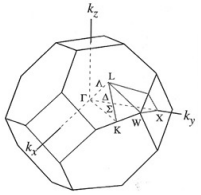


Complete Ionization

- When $E_d - E_F \gg kT$
  - $\frac{n_d}{N_d} = \frac{1}{1+\frac{1}{g}\exp(\frac{E_d-E_F}{kT})} \approx g\exp(\frac{-E_d+E_F}{kT})$
  - Total electron density: $n_0 = N_c \exp\left[\frac{-(E_c-E_F)}{kT}\right]$
  - Relative number of electrons in the donor state compared to the total number of electron:
    - ✓ $\frac{n_d}{n_d+n_0} = \frac{1}{1+\frac{N_c}{2N_d}\exp[\frac{-(E_c-E_d)}{kT}]}$
    - ✓ For example: for silicon, N_c=2.8e19 cm^{-3}, T=300K, phosphorous doping N_d = 1e16 cm^{-3}, Ionization energy is 0.045eV, what is percentage?

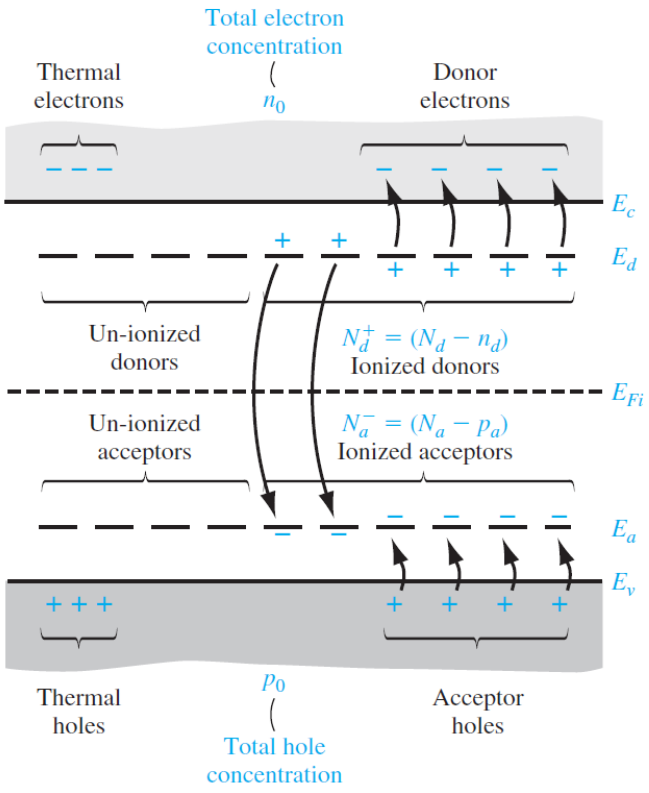


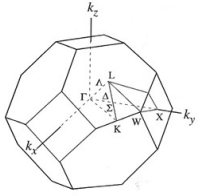
Charge Neutrality



Compensated Semiconductor

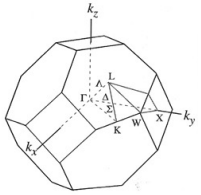
$$n_0 + N_a^- = p_0 + N_d^+$$





Equilibrium Electron and Hole Concentration

- $n_0 + N_a^- = p_0 + N_d^+$
- If complete ionization:
  - $n_0 + N_a = p_0 + N_d$
  - $n_0 + N_a = \frac{n_i^2}{n_0} + N_d$
    - ✓ $n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$ for majority carrier
    - ✓ $p_0 = \frac{n_i^2}{n_0}$ for minority carrier
    - ✓ $p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ for majority carrier
    - ✓ $n_0 = \frac{n_i^2}{p_0}$ for minority carrier



Electron Concentration Versus Temperature

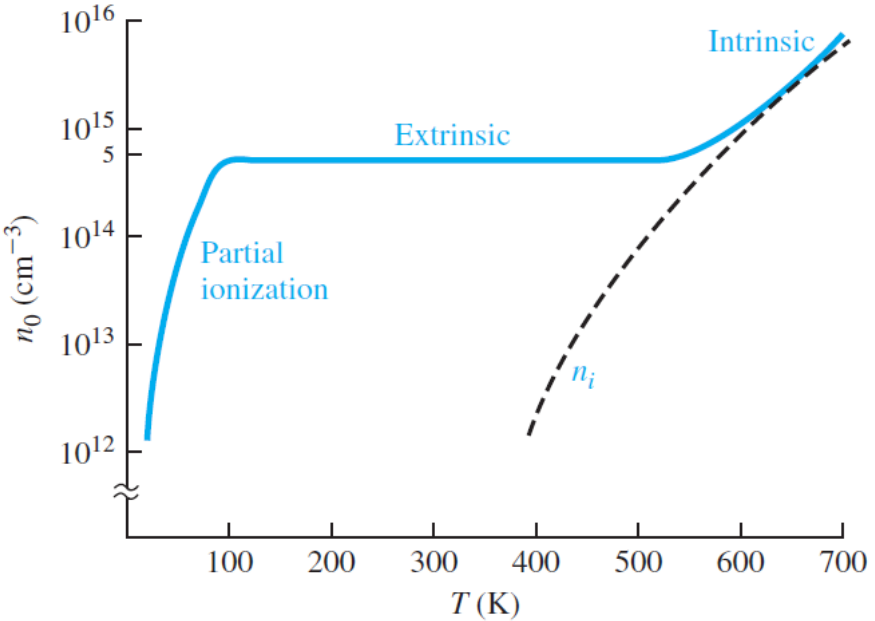
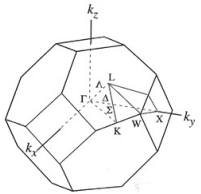
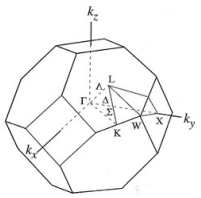


Figure 4.16 | Electron concentration versus temperature showing the three regions: partial ionization, extrinsic, and intrinsic.



Position of Fermi Level

- $n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$
  - $E_c - E_F = kT ln(\frac{N_c}{n_0}) \approx kT ln(\frac{N_c}{N_d})$
- $p_0 = N_v \exp\left[\frac{(E_v - E_F)}{kT}\right]$
  - $E_F - E_v = kT ln(\frac{N_v}{p_0}) \approx kT ln(\frac{N_v}{N_a})$
- $n_0 = n_i \exp\left[\frac{-(E_{Fi} - E_F)}{kT}\right]$
  - $E_F - E_{Fi} = kT ln(\frac{n_0}{n_i})$
- $p_0 = n_i \exp\left[\frac{(E_{Fi} - E_F)}{kT}\right]$
  - $E_{Fi} - E_F = kT ln(\frac{p_0}{n_i})$



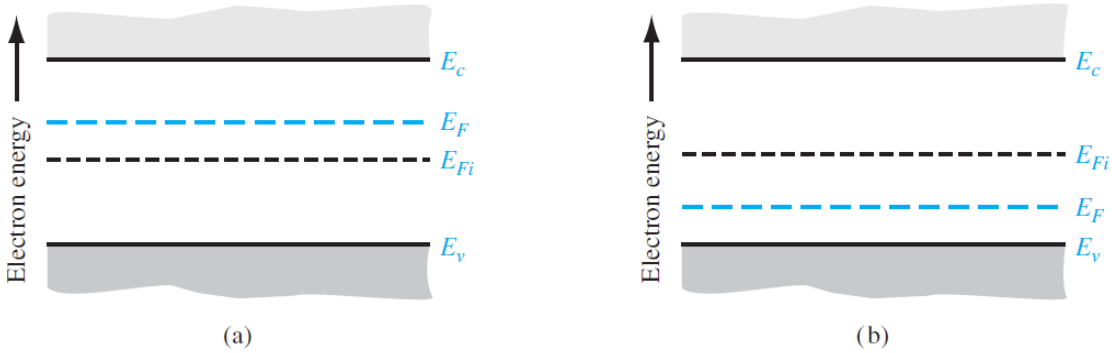
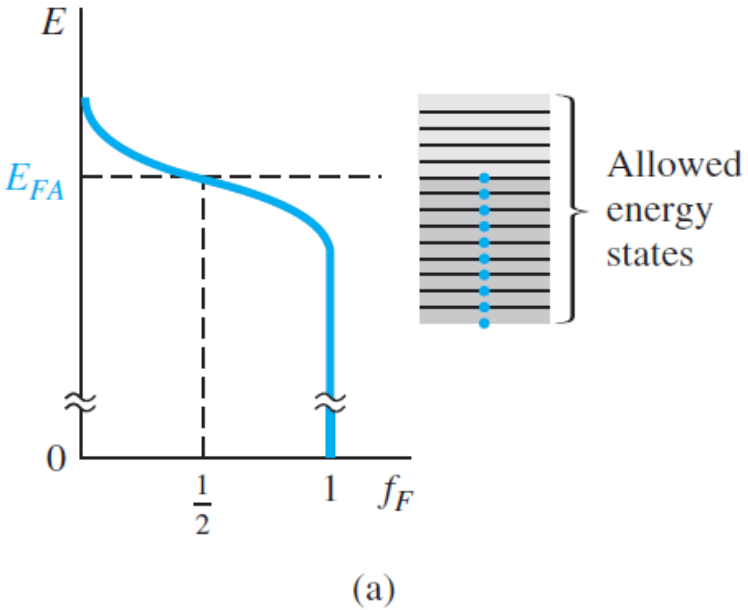
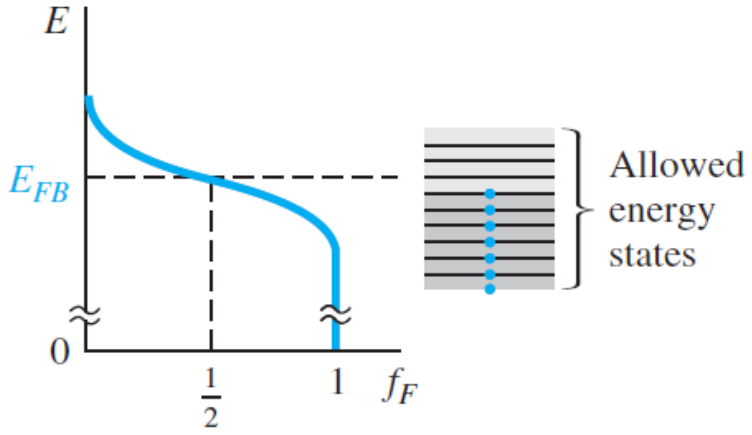


Figure 4.17 | Position of Fermi level for an (a) n-type ($N_d > N_a$) and (b) p-type ($N_d > N_a$) semiconductor.

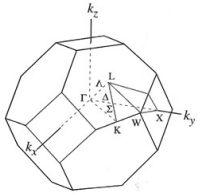
Relevance of the Fermi Level



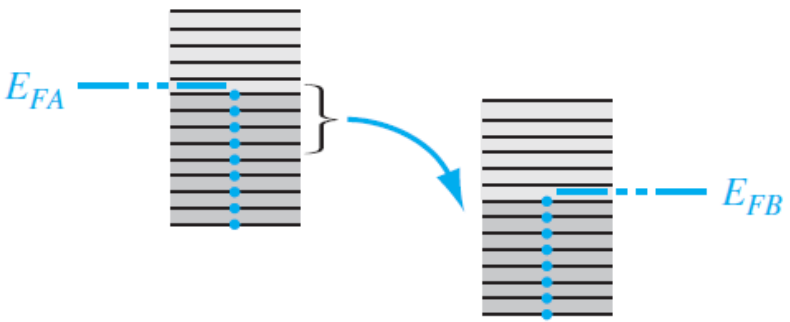
(a)



(b)



Relevance of the Fermi Level



(c)



(d)

